

*SELENE design map, companion to the BRONTE Chimborazo sheet (KER-BRO-TRK-001). One machine, one ridge: **EOS-3** (Ø3.0 m bore, Ø2.6 m pod, 7 t bulk per shot). Bore fixed across all five phases; only the track grows. Phase 1b baseline: 2.9 km at 100 g, escape speed 2.38 km/s, 8.7 MW south-pole solar. Total build estimate: \$2.0B–\$2.5B (mid-case, see Panel F). Mission one: cargo and satellite dispensers launched from the Moon and returned to Earth or delivered to cislunar orbit. The site map uses real NASA LRO/LOLA laser topography (20 m/px polar DEM); track alignment is conceptual. All physics re-derived June 2026.*

ANNEX A

SELENE, South Pole Design Map

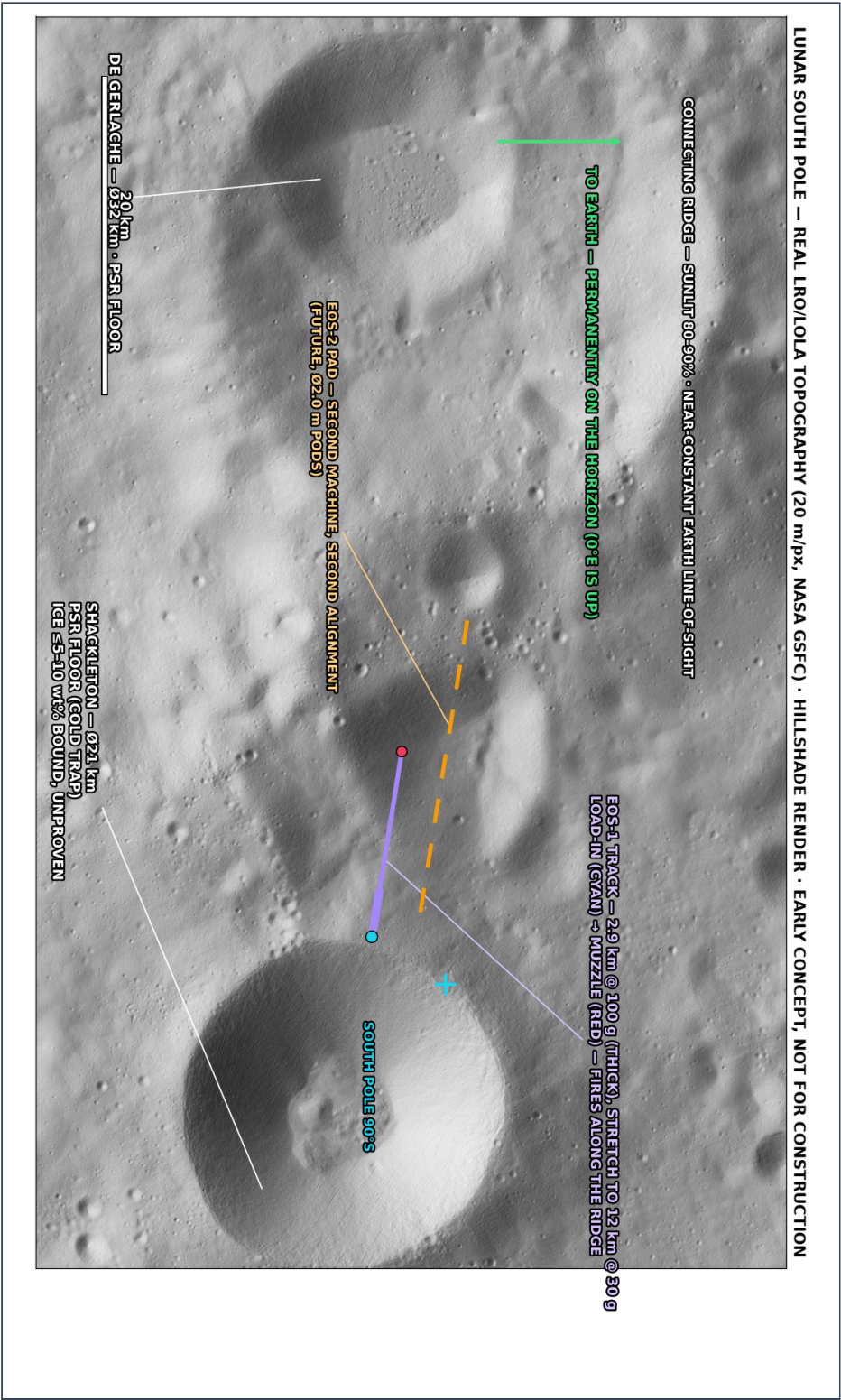
KER-SEL-TRK-002 • REV E • Ø3 m BORE • EARLY-VERSION TEST MOCK-UP • NOT FOR CONSTRUCTION

WHAT IS INSIDE

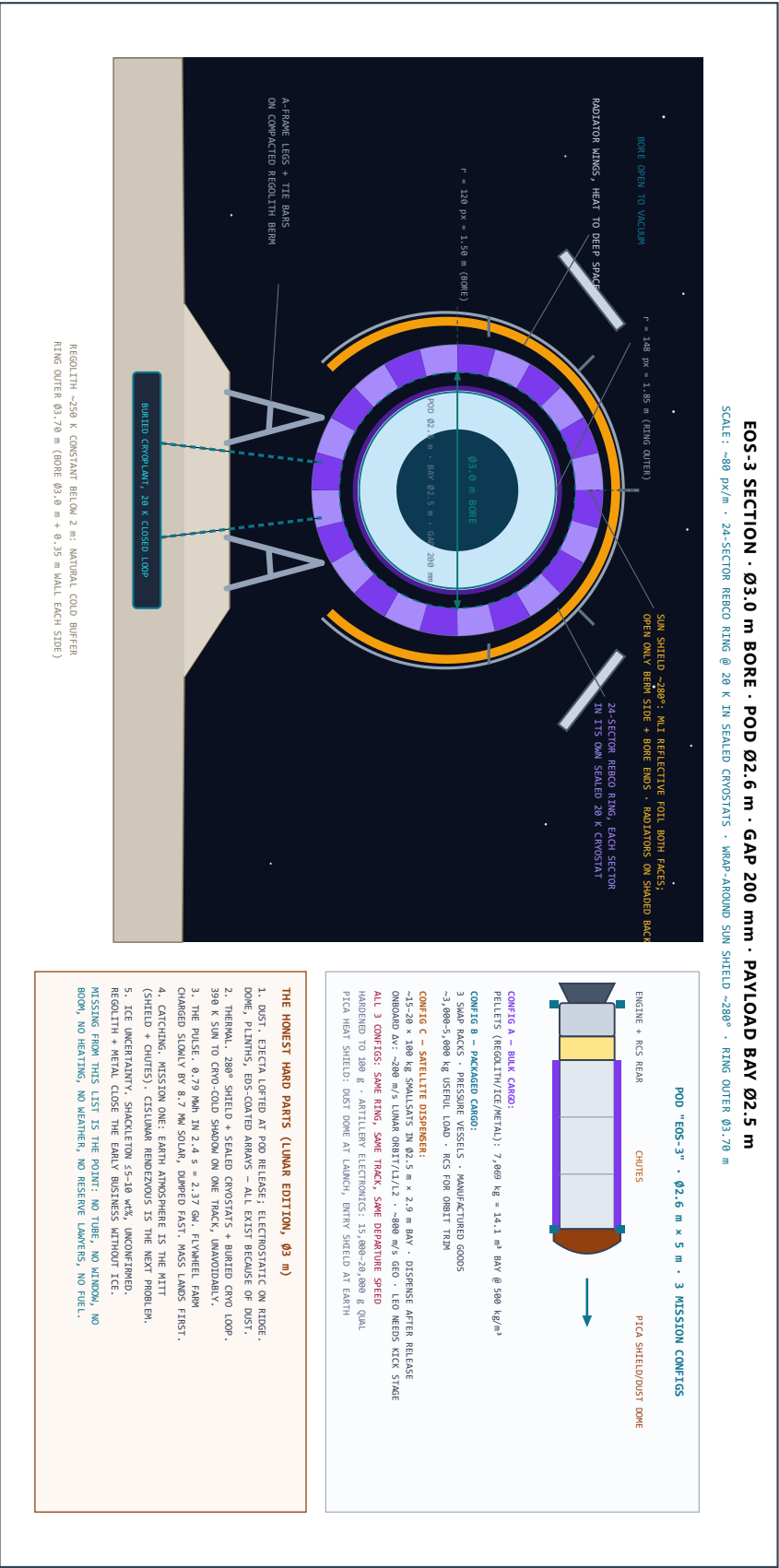
- ▶ Panel A, Site: real LRO/LOLA laser topography, Shackleton / Connecting Ridge ~89.7°S
- ▶ Panel A2, Where on the Moon · what does not exist here · EOS-3 machine specs (Ø3 m bore)
- ▶ Panel B, EOS-3 open-track section: no tube, the Moon is the vacuum vessel (Ø3.0 m bore)
- ▶ Panel C, Side profile: the raised track on its regolith embankment (true 1:1)
- ▶ Panel D, Engineering charts: track length vs g, Δv destinations, ridge illumination, energy
- ▶ Panel E, Return & delivery: the Earth–Moon system, Lagrange points, and how orbits actually work
- ▶ Panel F, Build cost summary (revised Ø3 m estimate, \$2.0B–\$2.5B mid-case)

Each large panel is on its own page, rotated to read at full size, turn the page (or your screen) a quarter-turn clockwise.

PANEL A, SITE: REAL LRO/LOLA TOPOGRAPHY, SHACKLETON / CONNECTING RIDGE, ~89.7°S



PANEL B, EOS-3 OPEN-TRACK SECTION, NO TUBE: THE MOON IS THE VACUUM VESSEL – Ø3.0 m BORE



POD "EOS-3" · Ø2.6 m × 5 m · 3 MISSION CONFIGS

ENGINE + RCS REAR CHUTES PICA SHIELD/DUST DOME

The diagram shows a side view of the EOS-3 pod. It is a cylindrical pod with a diameter of 2.6 meters and a length of 5 meters. The pod is divided into three main sections: the engine and RCS at the rear, the chutes in the middle, and the PICA shield/dust dome at the front. The pod is shown in a vertical orientation with an arrow pointing downwards, indicating the direction of travel.

CONFIG A – BULK CARGO:
PELLETS (REGOLITH/ICE/METAL): 7,000 kg = 14.1 m³ BAY @ 500 kg/m³

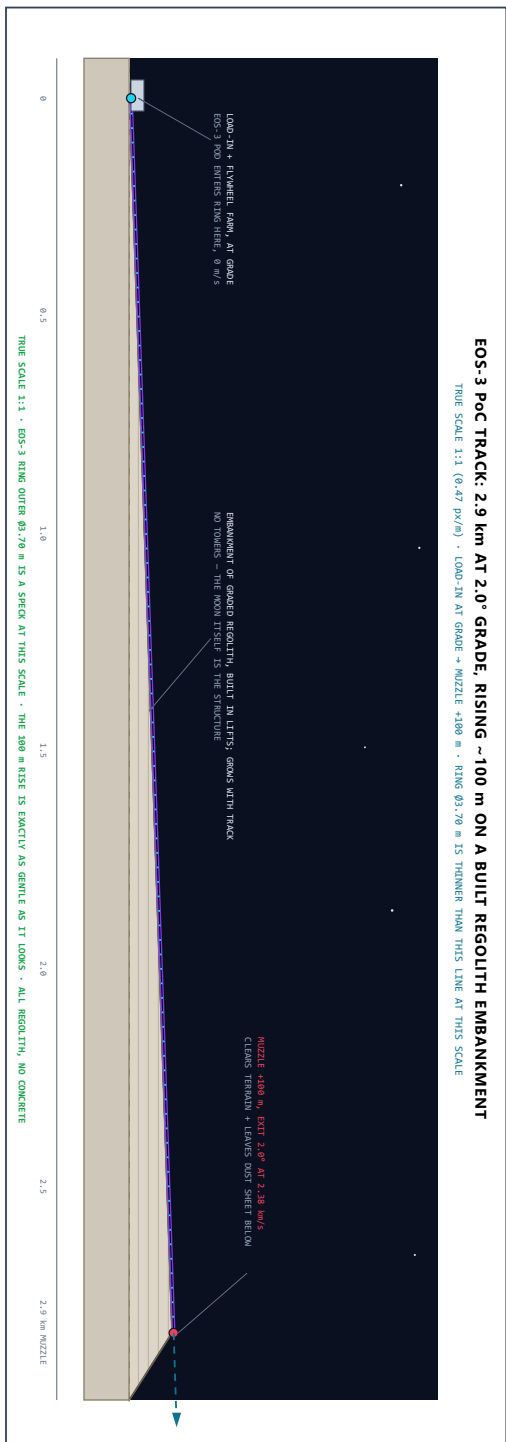
CONFIG B – PACKAGED CARGO:
3 SWAP RACKS · PRESSURE VESSELS · MANUFACTURED GOODS
~3,000-5,000 kg USEFUL LOAD · RCS FOR ORBIT TRIM

CONFIG C – SATELLITE DISPOSER:
~15-20 × 100 kg SATELLITES IN Ø0.5 m × 2.9 m BAY · DISPENSE AFTER RELEASE ORBIFORD DAY: ~200 m/s DOWN ORBIT/L/L2 · ~800 m/s GEO · LEO MEDS ROCK STAGE
ALL 3 CONFIGS: SAME RING, SAME TRACKS, SAME DEPARTURE SPEED
HARDENED TO 100 G · ARTILLERY ELECTRONICS: 15,000-20,000 G QUAL
PICA HEAT SHIELD: DUST DOME AT LAUNCH, ENTRY SHIELD AT EARTH

THE HONEST HARD PARTS (LUNAR EDITION, Ø3 m)

1. DUST, EJECTA LOFTED AT POD RELEASE; ELECTROSTATIC ON RIDGE, DOME, PLANTS, EOS-COATED ARMS – ALL EXIST BECAUSE OF DUST.
2. THERMAL, 280° SHIELD + SEALED CRYOSTATS + SEALED CRYO LOOP, 390 K SUN TO CRYO-COLD SHADOW ON ONE TRACK, UNAVOIDABLY.
3. THE PULSE, 0.79 MM IN 2.4 s = 2.37 GW, FLAMEBEL FARM CHARGED SLOWLY BY 8.7 MW SOLAR, DUMPED FAST. MASS LANDS FIRST.
4. CATCHING, MISSION ONE: EARTH ATMOSPHERE IS THE MITT (SHIELD + CHUTES). CISLUNAR RENDEZVOUS IS THE NEXT PROBLEM.
5. ICE UNCERTAINTY, SHACKLETON 55-10 W/3, UNCONFERMED. REGOLITH + METAL CLOSE THE EARLY BUSINESS WITHOUT ICE. MISSING FROM THIS LIST IS THE POINT: NO TUBE, NO WINDOW, NO BOOM, NO HEATING, NO WEATHER, NO RESERVE LAWYERS, NO FIU.

PANEL C, SIDE PROFILE: THE RAISED TRACK ON ITS EMBANKMENT (TRUE SCALE 1:1)



PANEL A2, WHERE ON THE MOON • WHAT DOES NOT EXIST • EOS-3 MACHINE SPECS

WHERE ON THE MOON?



PHOTO: REAL NEARSIDE • MAP: REAL LRO/LOLA LASER DATA
SHACKLETON RIM + CONNECTING RIDGE: 88-92% ANNUAL SUN

WHAT DOES NOT EXIST ON THIS SHEET

NO VACUUM TUBE — THE TRACK IS ALREADY IN HARD VACUUM
NO PLASMA WINDOW — NO ATMOSPHERE TO KEEP OUT
NO AERO-SHELL — NO NEEDLE NOSE — NO AIR TO CUT THROUGH
NO ISOLATION GATES — NO BLAST SHUTTERS — NO TUBE BREACH
NO SONIC BOOM — NO LAUNCH HEATING — NO DRAG, EVER

EVERY HARD SUBSYSTEM ON THE BRONTE SHEET IS DELETED BY ONE SENTENCE: THE MOON HAS NO ATMOSPHERE.
WHAT REMAINS IS THE RING, THE POWER, AND THE COLD.

WHAT REPLACES THEM: DUST + 250-390 K THERMAL SWINGS + GW-CLASS PULSE STORAGE + CATCHING THE CARGO AT THE OTHER END.

SITE SERVICES: RIDGE SOLAR 80-92% UPTIME • PSR COLD SINK NEXT DOOR • REGOLITH ~250 K AT 2 m DEPTH, NATURAL CRYO BUFFER

MAP DATA: NASA LRO/LOLA POLAR DEM, 20 m/px (LDEM_8755_20M), POLAR STEREO, RENDERED AS HILLSHADE. A REAL SURVEY.

CHOSEN BORE: Ø3 m. SAME RING ACROSS ALL 5 PHASES.
TRACK GRONIS; BORE DOES NOT CHANGE.

EOS-3 • Ø3 m BORE • THE CHOSEN MACHINE

PHASE 1b PoC: 2.9 km • POD Ø2.6 m • BORE Ø3.0 m • GAP 200 mm
7,069 kg BULK PER SHOT (14.1 m³ BAY @ 500 kg/m³) • 3,000-5,000 kg CARGO
ENERGY 0.79 MWh/SHOT • PEAK ~2.37 GW • RECHARGE ~5 min @ 8.7 MW
~100 SHOTS/HR BULK OR ~10 LARGE CARGO SHOTS/HR

THROUGHPUT: ~72,000 t/yr AT 8.7 MW (POWER-LIMITED, NOT BORE-LIMITED)
THROUGHPUT RULE: 8,378 t/MWh/yr AT ESCAPE SPEED
SAME TONNAGE ALL BORE SIZES: BORE BUYS SHOT MASS, NOT ANNUAL MASS

REBCO: 2,809 km TAPE • 6.7 t • \$281M @\$100/m • \$13M DELIVERY
24-SECTOR RING @ 20 K, 0.6 m PITCH • 4,833 RINGS OVER 2.9 km
ALL 5 PHASES USE IDENTICAL RING: TRACK EXTENDS, RING STAYS

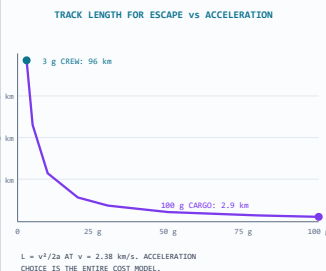
SCALING LADDER: 2.9 km → 12 km → 30 km → 50 km → 96 km
ESCAPE (2.38) • CARGO+ (2.66) • HIGH-SPD (4.20) • LLO/3g • ESC/3g

FIRST BILL IS A ROCKET BILL: 6.7 t REBCO + ELECTRONICS + SOLAR PLANT
CO-MANIFESTED ON 2-3 STARSHIP LANDER FLIGHTS • ~\$254M DELIVERY

RETURN MISSIONS: CARGO, SAMPLES, SATELLITE DISPENSERS (SEE PANEL E)
POD'S PICA HEAT SHIELD DOUBLES AS DUST DOME AT LAUNCH AND ENTRY SHIELD FOR DIRECT ATMOSPHERIC RECOVERY AT EARTH.
BUILD ESTIMATE: \$2.00-\$2.50 MID-CASE (SEE PANEL F)

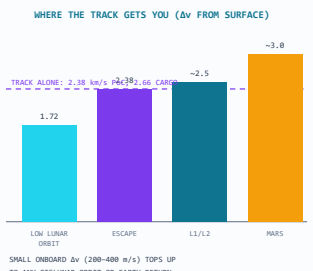
PANEL D, ENGINEERING CHARTS: TRACK LENGTH • DESTINATIONS • ILLUMINATION • ENERGY

TRACK LENGTH FOR ESCAPE vs ACCELERATION



L = v²/2a AT v = 2.38 km/s. ACCELERATION CHOICE IS THE ENTIRE COST MODEL.

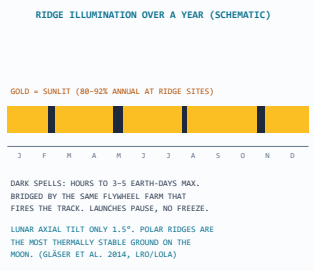
WHERE THE TRACK GETS YOU (Δv FROM SURFACE)



TRACK ALONE: 2.38 km/s PHASE CARGO: 3.8

SMALL ONBOARD Δv (200-400 m/s) TOPS UP TO ANY CISLUNAR ORBIT OR EARTH RETURN.

RIDGE ILLUMINATION OVER A YEAR (SCHEMATIC)

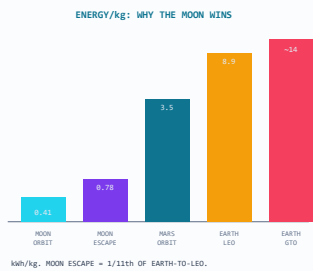


GOLD = SUNLIT (80-92% ANNUAL AT RIDGE SITES)

DARK SPELLS: HOURS TO 3-5 EARTH-DAYS MAX. BRIDGED BY THE SAME FLYWHEEL FARM THAT FIRES THE TRACK. LAUNCHES PAUSE, NO FREEZE.

LUNAR AXIAL TILT ONLY 1.5°. POLAR RIDGES ARE THE MOST THERMALLY STABLE GROUND ON THE MOON. (GLÄSER ET AL. 2014, LRO/LOLA)

ENERGY/kg: WHY THE MOON WINS



kWh/kg. MOON ESCAPE = 1/11th OF EARTH-TO-LEO.

EVERY DEPOT, STATION, OR 'SHEPPARD' GETS CHEAPER MASS FROM THE MOON THAN FROM EARTH.

PANEL E, RETURN & DELIVERY: THE EARTH-MOON SYSTEM AND HOW ORBITS ACTUALLY WORK

THE EARTH-MOON SYSTEM: LAUNCH DIRECTION, NOT JUST SPEED, SETS THE DESTINATION

EM-L1 lies ~58,000 km TOWARD Earth • EM-L2 lies ~62,000 km BEYOND the far side • the Moon orbits Earth at 1.02 km/s — geometry to scale, body sizes not



THE PRINCIPLE, DIRECTION NOT JUST SPEED

Escape speed (2.38 km/s) alone just leaves the payload in the Moon's own orbit around Earth. What you do relative to the Moon's 1.02 km/s motion decides everything: launch RETROGRADE to drop perigee toward Earth; launch PROGRADE / faster to climb to L-points, GEO, or escape.

RETURN TO EARTH SURFACE

samples - He-3 • ZBLAN fiber • pharma crystals

- Launch retrograde to the Moon's motion, ~2.5 km/s
- Coast a trans-Earth ellipse: apogee at the Moon, perigee grazing the atmosphere
- 3-5 day transit; arrive ~11 km/s (Apollo-class)
- PICA shield + parachutes, ocean / land recovery

A prograde shot would miss Earth and climb to a higher orbit instead — direction is the whole trick.

CISLUNAR ORBITS (satellites & relays)

~15-20 × 100 kg ESPA-class smallsats per shot (s7 t)

- Low lunar orbit: circularize at apolune (~0.3 km/s)
- EM-L1 halo, 58,000 km on the Earth-facing side
- EM-L2 / NRHO, 62,000 km past the far side (Gateway)
- Each sat adds its own small Δv to fine-tune

GEO transfer ~0.8 km/s; Earth LEO needs a ~3.1 km/s kick stage — it arrives at lunar-return speed, not orbital.

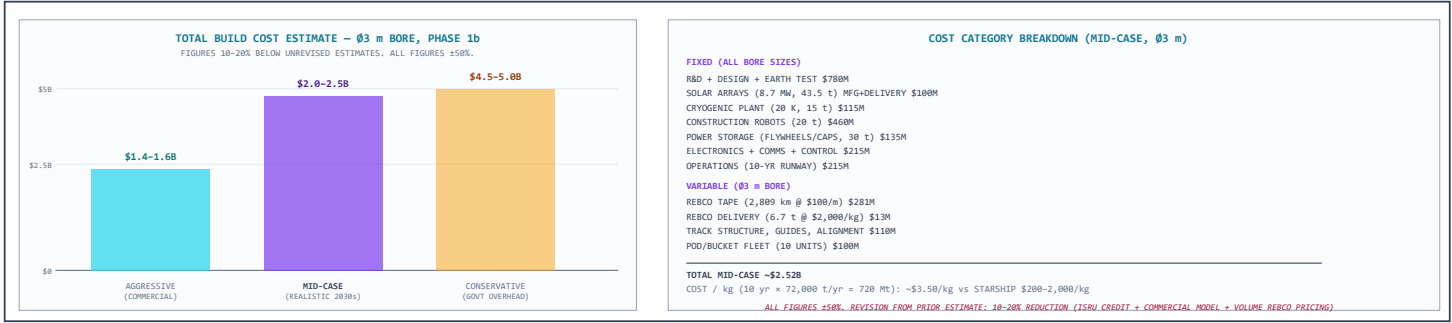
CISLUNAR DEPOTS (propellant, water, metal)

LOX/LH₂ - water - regolith shielding - metal ingots

- Aim the launch so apolune meets the depot's orbit
- Insertion / rendezvous burn ~150-300 m/s
- Berth or net-capture at an LLO or NRHO depot
- Transfer cargo: the pod flies back to reload

Delivered LOX / water ~\$3.50/kg vs \$3,000+/kg from Earth — this is the trade that funds the cislunar economy.

PANEL F, BUILD COST SUMMARY (REVISED ESTIMATE, Ø3 m, PHASE 1b)



SELENE DESIGN MAP (KER-SEL-TRK-002 REV E) · Ø3 m BORE · PHASE 1b PoC (2.9 km, 8.7 MW, 100 g, ESCAPE) · All physics derived June 2026, all hardware dimensions and cost preliminary. Real LRO/LOLA topography (NASA PDS, LDEM_875S_20M). © 2026 Perpetual Technologies. All rights reserved. Not for construction.